AP CALCULUS AB

Unit 1 • Lesson 1.1 practice • Answer key

Lesson 1.1 • Introducing Calculus: Can Change Occur at an Instant?

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

For $s(t) = t^2$, which expression gives the average rate of change on $[2, 2+h]$, $h \neq 0$?

(A) $4 + h$
(B) 4
(C) $2 + h$
(D) $4h + h^2$

Solution

Answer: (A). $\dfrac{(2+h)^2 - 4}{h} = \dfrac{4h + h^2}{h} = 4 + h.$

Question 2

Using the previous question, what is the instantaneous rate of change of s at $t = 2$?

(A) 2
(B) 4
(C) 0
(D) It cannot be found.

Solution

Answer: (B). As $h \to 0$, $4 + h \to 4$.

Question 3

The average velocities of an object on $[3, 3+h]$ are 6.1, 6.01, 6.001 for $h = 0.1, 0.01, 0.001$. Which is the best estimate of its velocity at $t = 3$?

(A) 6.1
(B) 6.001
(C) 6
(D) 3

Solution

Answer: (C). The average velocities approach 6 as h approaches 0.

Question 4

What is the instantaneous rate of change of $f(x) = 3x + 2$ at $x = 5$?

(A) 17
(B) 5
(C) 2
(D) 3

Solution

Answer: (D). A linear function has the same rate of change, its slope, at every point.

Question 5 (Free response)

An object's position is $s(t) = t^3$ meters at time t seconds.

(a) Show that the average velocity on $[1, 1+h]$ is $3 + 3h + h^2$.
(b) Compute it for $h = 0.1$ and $h = 0.01$.

(c) Estimate the velocity of the object at $t = 1$, with units.

Solution

(a) $\dfrac{(1+h)^3 - 1}{h} = \dfrac{1 + 3h + 3h^2 + h^3 - 1}{h} = 3 + 3h + h^2$.

(b) $h = 0.1$: 3.31 m/s. $h = 0.01$: 3.0301 m/s.

(c) As $h \to 0$ the average velocity approaches 3: the velocity at $t = 1$ is 3 meters per second.